EXPERIMENT NO: 10

Experiment Title: Load and display JSON data using fetch() and jQuery $.getJSON(); render results dynamically in a table.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

JSON (JavaScript Object Notation): JSON is a lightweight data-interchange format used to store and exchange data between web applications and servers. It represents data using key-value pairs and arrays, making it easy to process using JavaScript.

Fetching JSON Data: JavaScript provides the fetch() API to request data from a server. The response can be converted into a JavaScript object using the .json() method. The then() and catch() methods handle successful responses and errors.

jQuery $.getJSON() Method: jQuery provides the $.getJSON() method to retrieve JSON data from a server. It simplifies AJAX requests and allows developers to process the returned data using a callback function.

Key Methods and Functions:

1. fetch(): Sends a request to retrieve data from a specified URL.
2. response.json(): Converts the response body into JavaScript data.
3. $.getJSON(): Loads JSON data asynchronously using jQuery.
4. DOM Manipulation: JavaScript dynamically creates and updates HTML table rows using methods such as document.createElement() and appendChild().
5. Error Handling: The catch() method and jQuery's failure callback handle errors that occur while retrieving data.

Dynamic Table Rendering: In this experiment, JSON data containing user information is retrieved using both fetch() and jQuery $.getJSON(). The retrieved records are displayed dynamically in an HTML table containing fields such as ID, Name, Email, and City. This demonstrates how asynchronous data retrieval and DOM manipulation can be used to display server data without reloading the entire webpage.

Experiment Program Code

Code 1: jquery.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <script src = "https://code.jquery.com/jquery-3.7.1.min.js"></script>
</head>
<body>
```

```
    <h2>My First jQuery Program</h2>
    <p id = "demo">Welcome to jQuery!</p>
    <button id="btn">Hide Paragraph</button>

    <script>
        $(document).ready(function(){
            $("#btn").click(function(){
                $("#demo").hide();
            });
        });
    </script>
</body>
</html>
```

5. Case Study Title

Load and display JSON data using fetch() and jQuery $.getJSON(); render results dynamically in a table.

6. Case Study Program Code

Casestudy_10_Kanak_Tembhurne_1290.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Experiment 10 - JSON Data</title>

<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

<style>
body {
    font-family: Arial, sans-serif;
    background: #f1f5f9;
    padding: 20px;
}
.container {
    max-width: 800px;
    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 10px;
}
h2 { text-align: center; color: #2563eb; }
button {
    padding: 10px;
    margin: 5px;
    background: #2563eb;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}
th, td {
    border: 1px solid #aaa;
    padding: 10px;
    text-align: left;
}
th { background: #2563eb; color: white; }
#status { margin-top: 15px; }
</style>
</head>
```

```
<body>
<div class="container">
    <h2>Load JSON Data</h2>

    <button onclick="useFetch()">Load using fetch()</button>
    <button onclick="useJQuery()">Load using $.getJSON()</button>

    <p id="status">Click any button to display data.</p>

    <table>
        <thead>
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Email</th>
                <th>City</th>
            </tr>
        </thead>
        <tbody id="result"></tbody>
    </table>
</div>

<script>
const users = [
    {
        id: 1,
        name: "Rahul Sharma",
        email: "rahul@gmail.com",
        city: "Pune"
    },
    {
        id: 2,
        name: "Priya Patil",
        email: "priya@gmail.com",
        city: "Mumbai"
    },
    {
        id: 3,
        name: "Amit Verma",
        email: "amit@gmail.com",
        city: "Delhi"
    }
];

// Convert sample data to JSON
const jsonData = JSON.stringify(users);

function displayData(data) {
    const tbody = document.getElementById("result");
    tbody.replaceChildren();

    data.forEach(user => {
        const row = document.createElement("tr");

        [user.id, user.name, user.email, user.city]
            .forEach(value => {
                const cell = document.createElement("td");
                cell.textContent = value;
                row.appendChild(cell);
            });

        tbody.appendChild(row);
    });
}

// Method 1: Fetch API
function useFetch() {
    const status = document.getElementById("status");

    fetch("data:application/json," + encodeURIComponent(jsonData))
        .then(response => response.json())
        .then(data => {
            displayData(data);
            status.textContent = "Data loaded using fetch()!";
        })
        .catch(error => {
```

```
            status.textContent = "Error: " + error.message;
        });
}

// Method 2: jQuery $.getJSON()
function useJQuery() {
    const status = document.getElementById("status");

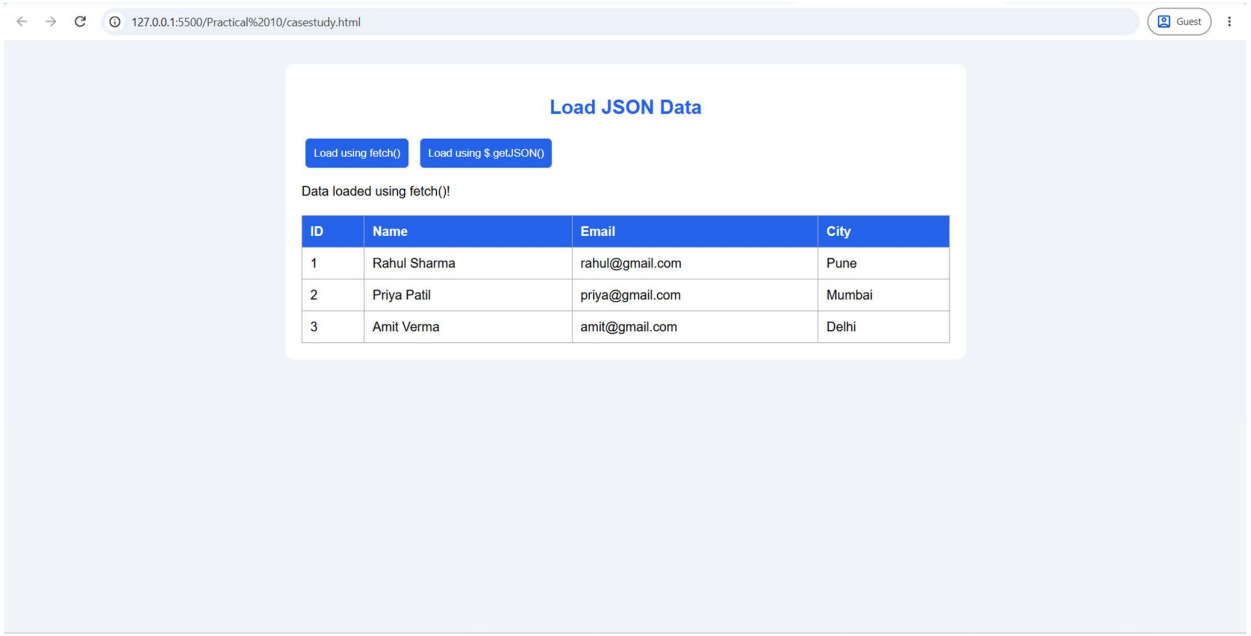
    if (!window.jQuery) {
        status.textContent =
            "jQuery did not load. Connect to the internet and refresh.";
        return;
    }

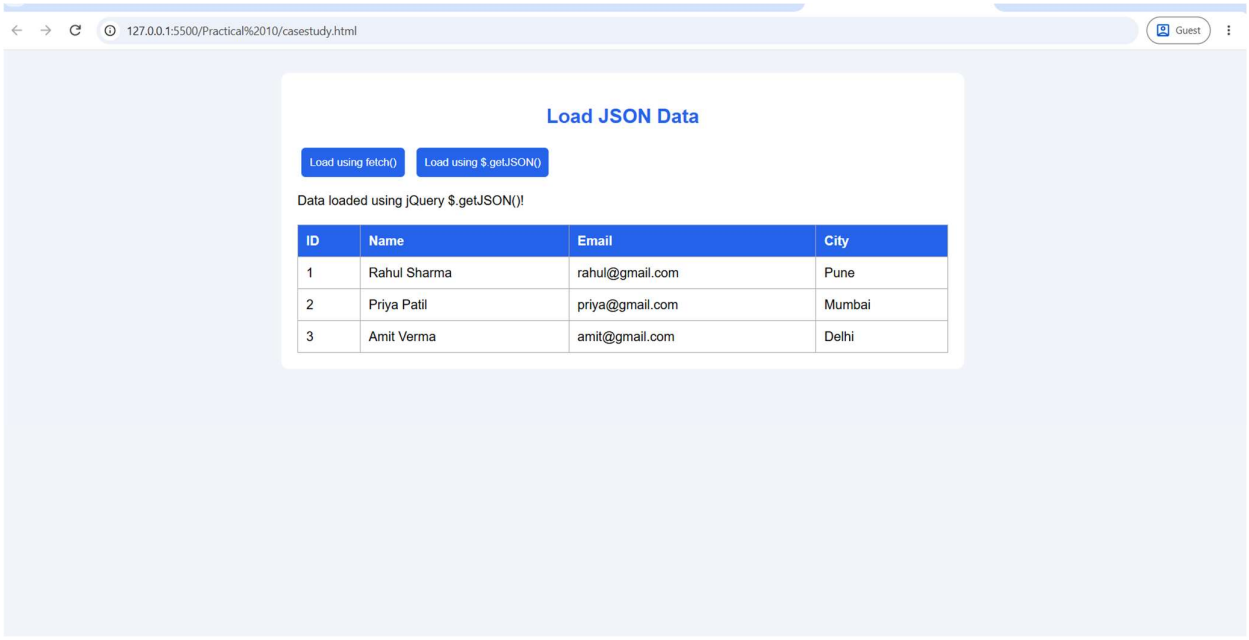
    const dataURL =
        "data:application/json," + encodeURIComponent(jsonData);

    $.getJSON(dataURL)
        .done(function(data) {
            displayData(data);
            status.textContent = "Data loaded using jQuery $.getJSON()!";
        })
        .fail(function() {
            status.textContent =
                "This browser blocked the data URL. Use the local-server method below.";
        });
}
</script>
</body>
</html>
```

7. Output of the case study :

Name : Kanak Tembhurne PRN : 24070521290 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_10_Kanak_Tembhurne_1290.html





8. Conclusion

Thus, JSON data was successfully retrieved using JavaScript fetch() and jQuery $.getJSON(), and the retrieved records were dynamically displayed in an HTML table using DOM manipulation.